

Divisibility Check

Day 84
02/10/2025

Check Whether Large Number is divisible by 12

Approach : This is Very Simple approach, if a Number is divisible By 4 and 3 then the Number is divisible By 12

Point 1 $\Rightarrow$ if the last two digits of the number are divisible by 4 then the number is divisible by 4

Point 2 $\Rightarrow$ if the sum of all digits of a number is divided by 3 then the number is divisible by 3

```
static boolean isDvisibleBy12(String num)
{
```

```
    // if number greater than 3
```

```
    if (num.length() >= 3) {
```

```
        // find last digit
```

```
        int d1 = (int)num.charAt(num.length() - 1);
```

```
        // no is odd
```

```
        if (d1 % 2 != 0)
```

```
            return false;
```

```
        // find second last digit
```

```
        int d2 = (int)num.charAt(num.length() - 2);
```

```
        // find sum of all digits
```

```
        int sum = 0;
```

```
        for (int i = 0; i < num.length(); i++)
```

```
            sum += num.charAt(i);
```

```
        return (sum % 3 == 0 &&
```

```
                (d2 * 10 + d1) % 4 == 0);
```

```
    }
```

```
    else {
```

```
        // if number is less than
```

```
        // or equal to 100
```

```
        int number = Integer.parseInt(num);
```

```
        return (number % 12 == 0);
```

```
    }
```

```
    // driver function
```

```
}
```

$\rightarrow$ if num length is greater than 2

$\rightarrow$ Extract last digit

$\rightarrow$ if the number is odd then it's Not divisible by 12

$\rightarrow$ extract second last digit

$\rightarrow$ Divisibility check 3 & 4

```
public static void main (String[] args) {
```

```
    String num = "12244824607284961224";
```

```
    if (isDvisibleBy12(num))
```

```
        System.out.print("Yes");
```

```
    else
```

```
        System.out.print("No");
```

```
}
```

check if a Large Number is divisible By 13 or not

Input: $s = "2911285"$

Output: true

Explanation: $2911285 / 13 = 223945$, which is a whole number with no remainder.

Input: $s = "27"$

Output: false

Explanation: $27 / 13 \approx 2.0769...$, which is not a whole number (there is a remainder).

Approach: A number is divisible by 13 if and only if the Alternating Sum of its 3-digit blocks, taken from right to left, is divisible by 13.

// Function to check divisibility by 13

```
static boolean divBy13(String s) {  
    int len = s.length();
```

```
// Special case: if the number is "0"  
if (len == 1 && s.charAt(0) == '0') {  
    return true;  
}
```

```
// Make the length a multiple of 3 by padding zeros at the end
```

```
if (len % 3 == 1) {  
    s += "00";  
    len += 2;  
} else if (len % 3 == 2) {  
    s += "0";  
    len += 1;  
}
```

```
int sum = 0;  
int p = 1;
```

```
// Traverse from right to left in steps of 3 digits
```

```
for (int i = len - 1; i >= 0; i--) {  
    int group = 0;  
    group += s.charAt(i--) - '0';  
    group += (s.charAt(i--) - '0') * 10;  
    group += (s.charAt(i) - '0') * 100;
```

```
    sum += group * p;  
    p *= -1;  
}
```

```
sum = Math.abs(sum);  
return sum % 13 == 0;
```

Step by Step approach -

- Pad the number so its length is a multiple of 3
-> If the number of digits is not a multiple of 3, append zeros to the right so each block has exactly 3 digits.
-> Example: "2911285" → "291128500" (after padding with two zeros).
- Split into 3-digit blocks from right to left
-> Example: "291128500" → blocks: 500, 128, 291 (right to left order).
- Apply alternating signs starting with + on the rightmost block
-> Pattern from right to left: + block, - block, + block, ...
-> Example: $+500 - 128 + 291$.
- Sum the results
-> Example: $500 - 128 + 291 = 663$.
- Check divisibility by 13
-> If the sum is divisible by 13, the original number is divisible by 13.
-> Example: $663 \% 13 == 0 \rightarrow$ divisible.

→ Alternate 3 blocks sum left to right

```
public static void main(String[] args) {  
    String s = "2911285";  
    System.out.println(divBy13(s));  
}
```

Approach 2: String Based Modulo

We process the number digit by digit from left to right, maintaining the remainder modulo 13 at each step using the formula

Step by Step Approach -

- Initialize remainder:
-> $rem = 0$
- Process each digit from left to right:
-> Digit '2': $rem = (0 * 10 + 2) \% 13 = 2$
-> Digit '9': $rem = (2 * 10 + 9) \% 13 = 29 \% 13 = 3$
-> Digit '1': $rem = (3 * 10 + 1) \% 13 = 31 \% 13 = 5$
-> Digit '1': $rem = (5 * 10 + 1) \% 13 = 51 \% 13 = 12$
-> Digit '2': $rem = (12 * 10 + 2) \% 13 = 122 \% 13 = 6$
-> Digit '8': $rem = (6 * 10 + 8) \% 13 = 68 \% 13 = 5$
-> Digit '5': $rem = (5 * 10 + 5) \% 13 = 55 \% 13 = 0$
- Since final $rem = 0$, the number 2911285 is divisible by 13.

```
static boolean divBy13(String s) {  
    // Stores running remainder  
    int rem = 0;  
  
    // Process each digit and compute remainder modulo 13  
    for (int i = 0; i < s.length(); i++) {  
        rem = (rem * 10 + (s.charAt(i) - '0')) % 13;  
    }  
  
    // Final check: if remainder is 0, number is divisible by 13  
    return rem == 0;  
}  
  
public static void main(String[] args) {  
    String s = "2911285";  
  
    if (divBy13(s))  
        System.out.println("true");  
    else  
        System.out.println("false");  
}
```

Check if a large number is divisible by 15

Input: "31"

Output: No

Input : num = "156457463274623847239840239
402394085458848462385346236
482374823647643742374523747
264723762374620"

Output: Yes

Given number is divisible by 15

A number is divisible by 15 if it is divisible by 5 (if the last digit is 5 or 0), and it is divisible by 3 (if sum of digits is divisible by 3)

```

// function to check if a large
// number is divisible by 15
public static boolean isDivisible(String S)
{
    // length of string
    int n = S.length();

    // check divisibility by 5
    if (S.charAt(n - 1) != '5' &&
        S.charAt(n - 1) != '0')
        return false;

    // Sum of digits
    int sum = 0;
    for(int i = 0; i < S.length(); i++)
        sum += (int)S.charAt(i);

    // if divisible by 3
    if(sum % 3 == 0)
        return true;
    else
        return false;
}

```

→ divisibility By 5

3 divisibility
check

```

public static void main (String[] args)
{
    String S = "15645746327462384723984023940239";
    if(isDivisible(S) == true)
        System.out.println("Yes");
    else
        System.out.println("No");

    String S1 = "15645746327462384723984023940235"
    if(isDivisible(S1) == true)
        System.out.println("Yes");
    else
        System.out.println("No");
}

```

Number is divisible By 29 or not

Input : 363927598

Output : No

Input : 292929002929

Output : Yes

```

// Returns true if n is divisible by 29
// else returns false.
static boolean isDivisible(long n)
{
    // add the lastdigit*3 to renaming
    // number until number comes only
    // 2 digit
    while (n / 100 > 0) {

        int last_digit = (int)n % 10;
        n /= 10;
        n += last_digit * 3;
    }

    // return true if number is
    // divisible by 29 another
    return (n % 29 == 0);
}

```

a quick solution to check if a number is divisible by 29 or not is to add 3 times of last digit to next number and repeat this process until number comes 2 digit. The given number is 348.

Three times of last digit +
Rest of the Number = $34 + 8 \times 3 = 58$
Since 58 is divisible by 29, 348
is also divisible by 29.

```

// Driver code
public static void main(String[] args)
{
    long n = 348;

    if (isDivisible(n))
        System.out.println("Yes");
    else
        System.out.println("No");
}

```